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Weird Mechanics

Sarthak Arora[1]

Abstract
This article provides a brief introduction to the counter-intuitive principles of Quantum Mechanics, designed for those with an interest in physics. We, very briefly, discuss the basic mathematical foundation required to properly formulate the theory, along with the idea behind Quantum Tunneling and the application of it in the real world. The aim is to show that seemingly abstract quantum concepts have tangible and powerful applications in the modern world.

[1] *Indian Institute of Science, Bengaluru, India*
✉ aroraa.ssahil@gmail.com

Introduction

The possibility of a person, interested in physics, not coming across the term "Quantum" is just negligible. In fact, Quantum Mechanics is infamous for its counter-intuitive results. A full-fledged development of the theory of Quantum Mechanics started in 1925, which is why we celebrated *100 years of Quantum* this year. The purpose of this article would be to give a slight notion of what Quantum Mechanics is about and its applications in today's world.

1. What is Quantum Mechanics?

Imagine spending years in school to understand classical physics and then in university you find ample number of examples crushing your all the intuition you developed. This is what actually happens to many students when they come across this topic. Let me begin with with a question. If I have a ball, having energy E (consider only kinetic and potential energy), and it tries to enter a region in which the potential energy would become higher than E, is it really possible? This may seem like a simple question and the most probable answer would be a big NO, but this is actually what happens in Quantum Mechanics. In fact, this is *exactly* what happens in nuclear fusion in stars, but we will come to that later. The main thing that differentiates Quantum Mechanics from Classical Mechanics is that we talk about a quantity known as the *Wavefunction*. Now, we have several interpretations about this wavefunction. Some interpretations, like the Born Interpretation, take the square of this wavefunction to be the probability (density). There is another interpretation, a quite famous one, known as the *Many-World Interpretation* (MWI). This interpretation is quite controversial and was questioned by prominent physicists like Bohr and Heisenberg themselves. Due to its highly complex nature as compared to Copenhagen Interpretation (linked to the wavefunction collapse upon measurement, in contrast to Quantum Decoherence in MWI), we will not discuss it further.

The main takeaway from the previous paragraph was that to introduction of a new quantity *Wavefunction*, whose square (actually the square of the magnitude, because the wavefunction is generally complex) acts as the probability (density). Your next question should be, "*What are these wavefunctions? Rationals? Polynomials? Apples? Bananas?*". Well, wavefunctions can be taken as functions, but there is a more correct way to define it[1]. Wavefunctions are elements of a special space known as the *Hilbert Space*. The Hilbert Space has some specific properties that makes it special and a good choice to define wavefunctions. In the Modern world we denote the wavefunction or better to call, a state) as $|\psi\rangle$. The best thing about this formulation is that these states are vectors in the Hilbert Space. This enables us to define operators as matrices, which gives us the power to use our knowledge of Linear Algebra to study the theory of Quantum Mechanics.

2. Mathematics of Quantum Mechanics

As stated earlier, Linear Algebra plays a huge role in the theory of Quantum Mechanics. The following sections expect some mathematical maturity from the readers. I highly encourage to refer to [1] for some proper mathematical structure of Hilbert Space.

3. Quantum Tunneling

Now, let's try to analyze the process of Quantum Tunneling. For this case, we will work with functions instead of state

[1] There is even a more correct way where we define Manifolds as the base space, Fibers on the Manifold and then relating it to the wavefunction. I encourage the interested to look this up.

vectors, simply because they are easier to work with while solving Differential Equations. The derivations in this section are inspired from the lecture notes of David Tong [2]. Imagine you have a potential function given by Figure 1 ,

$$V(x) = \begin{cases} 0 & x < 0 \\ U & x > 0 \end{cases} \tag{1}$$

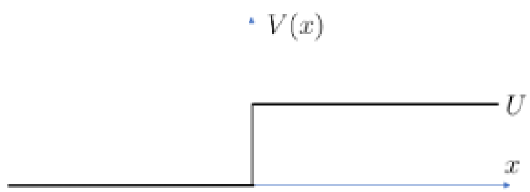

Figure 1. Potential Function V(x)

The value of U is such that $U > E$, where E is the total energy of the wave. Now, in the region $x < 0$, if we observe the Schrödinger equation,

$$-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi \tag{2}$$

Note that we have considered time independent version of Schrödinger equation because we are studying the variation of ψ only as a function of position (x). The energy, being positive, makes this differential equation similar to the Simple Harmonic case and the general solution would be $e^{\pm if(E)x}$, where $f(E)$ is a function of energy. This case gives us results that we normally expect but something weird happens when the wave enters the region $x > 0$. According to the Schrödinger equation,

$$-\frac{\hbar^2}{2m}\nabla^2\psi + U\psi = E\psi \tag{3}$$

$$\frac{\hbar^2}{2m}\nabla^2\psi = (U-E)\psi \tag{4}$$

Now, both side of the equation is positive so the general solution of this differential equation would be, $e^{\pm g(E)x}$. In fact, we will discard the case with the positive argument of the exponential to satisfy some boundary conditions, something that we are supposed to follow. This shows that the wavefunction exists in the region which is classically not allowed and the takes the form of exponentially decaying function with respect to the position. To make the above statement more rigorous, you can define the probability density in the region of positive x and then integrate to get a non-zero result, which shows that probability of finding the particle in that region, although can be very small, but, nonetheless, non-zero.

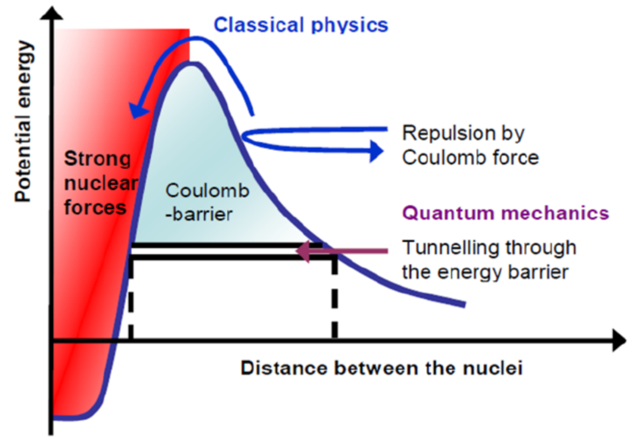

Figure 2. Coulombic Energy Barrier [3]

4. Applications of Quantum Mechanics

Whenever I tell such things to people, there is one common reply I get, "*Okay, but all this is just math, and since it doesn't make sense, this shouldn't happen, right?*", and probably you are thinking this too. The best thing about Quantum is that all this actually happens!

4.1 Nuclear Fusion in Stars

It is a very common knowledge that Stars generate power by the process of Nuclear Fusion. The main reaction happening in stars for the majority of their lifetime is the production of ^{4}He from ^{1}H, given by,

$$\begin{aligned} {}^1\mathrm{H} + {}^1\mathrm{H} &\rightarrow {}^2\mathrm{H} + e^+ + \nu_e \quad \text{(twice)} \\ {}^1\mathrm{H} + {}^2\mathrm{H} &\rightarrow {}^3\mathrm{He} + \gamma \quad \text{(twice)} \\ {}^3\mathrm{He} + {}^3\mathrm{He} &\rightarrow {}^4\mathrm{He} + {}^1\mathrm{H} + {}^1\mathrm{H} \end{aligned}$$

The interesting thing about this is that for the nuclear reactions to happen, the energy of the reactants has to be greater than the energy of Coulomb repulsion at the point of minimum distance, where the reactions happen. The only way reactants can gain this much energy is through thermal modes. If we assume this and try to calculate the temperature required at the core of our Sun for this reaction to occur, we get $T \sim 10^{25}K$, while we know, through techniques commonly used in Astrophysics, that actually $T \sim 10^7K$. It is when we use the Quantum Tunneling to cross the energy barrier of the Coulombic Repulsion, that we get the actual order of temperature.

Another application of Quantum, particularly in industry is in Quantum Computation.

4.2 Quantum Computation

Quantum Entanglement is, yet again, a very seemingly absurd phenomenon in Quantum Physics. Einstein called it as

"*Spooky action at a distance*". When particles become entangled, they form a single, unified quantum system, even if separated by vast distances. Their individual properties are no longer independent, instead, their fates are totally linked. The most counter-intuitive aspect of this connection is that the state of each particle is not definite until it is measured. Before measurement, they exist in a cloud of probabilities. However, the moment one particle is measured and its state "collapses" into a definite outcome, say, spin up, the state of its entangled partner is instantly determined to be the opposite, spin down, regardless of the separation between them. This influence is instantaneous, seemingly violating the cosmic speed limit of light. Note that this isn't about hidden information, like discovering one glove is left-handed and thus knowing the other is right-handed. In the quantum realm, the properties were not pre-determined, the act of measurement on one particle actively shapes the reality of the other.

This concept is the secret sauce of quantum computation. A classical computer uses bits, which are either a 0 or a 1. A quantum computer uses qubits, which, thanks to the principle of superposition, can be a 0, a 1, or both at the same time. While superposition provides a foundational advantage, entanglement is what scales this power exponentially. When you entangle qubits, you are no longer dealing with a collection of individual units but a single, powerful computational entity. Two entangled qubits, for instance, don't just represent two simultaneous states, but create a unified system that explores all four possible combinations (00, 01, 10, 11) in a correlated manner. Three entangled qubits explore eight combinations, and so on. For 'n' entangled qubits, the system can process and exist in a superposition of 2^n states at once. This exponential growth in computational space is something a classical computer could never achieve. The best example for this goes back to 2019, where Google's Syncamore quantum processor took only 200 seconds to solve a highly computational problem. When we estimate the time taken to solve the problem using the best classical computer, it comes around to be around 10,000 years.

In practice, quantum algorithms are designed specifically to manipulate these complex, entangled states. Quantum gates, the building blocks of a quantum circuit, don't just flip individual qubits, they create and modify the entanglement between them. It is this web that allows a quantum computer to perform massive parallel computations. By carefully preparing an entangled initial state and applying a sequence of quantum gates, the computer can explore a vast landscape of potential solutions simultaneously. The final measurement collapses this complex state into a single answer, which, due to the clever design of the algorithm, has a high probability of being the correct one.

5. Conclusion

So now you know that Quantum Mechanics isn't *just* math. Active work is going on to exploit this nature of Quantum Mechanics. Many tech projects are working to include Quantum into their systems and product, wherever possible. Quantum Mechanics isn't just confined to Physics. The probability of finding concepts of Quantum Mechanics in fields like Engineering, Biology and Chemistry is non-zero!

References

[1] David Skinner. Lectures on quantum field theory. Lecture notes, University of Cambridge. Available at `https://www.damtp.cam.ac.uk/user/dbs26/PQM/chap2.pdf`.

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[3] Cyriel Wagemans. *The Nuclear Fission Process*. CRC press, Boca Raton, FL, 1991.